DAY 17 | POST-READ

DevOps and Cloud: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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Part A: Summary of today

Topic	What to remember
DevOps	Shared ownership, small changes, automation, blameless learning.
DevOps loop	Plan, code, build and test, release, operate and monitor, repeat.
CI / delivery / deployment	Every push tested; every green build release-ready; or released automatically.
DevSecOps (NEW v4)	SAST (bandit), dependency scanning (pip-audit, npm audit), secret scanning in CI.
IaaS / PaaS / SaaS	Less to manage and less control as you move right.
Migration paths	Rehost, replatform, refactor, rearchitect, rebuild, replace.
Azure hierarchy	Tenant, subscription, resource group, resources; pick the region with care.
Azure DevOps	Boards, Repos, Pipelines, Test Plans, Artifacts.
Pipeline YAML	trigger, pool, stages, jobs and steps in azure-pipelines.yml.

Part B: Going beyond today

1. The shared responsibility model

Security is shared. In every model the cloud provider secures the physical data centres; you always remain responsible for your data, your users and how you grant access.

2. Triage of scan results

Not every finding is equal. Fix high-severity findings first, check whether the vulnerable code path is used, and record each decision. You started vulnerability report v1 on Day 7; Day 20 extends it to v2 with CI gates.

3. DORA metrics

Teams often track deployment frequency, lead time for changes, change failure rate and time to restore service. They show whether DevOps practices are working.

4. Infrastructure as Code

Instead of clicking in the portal, describe resources in files (Bicep, ARM or Terraform), review them like code and recreate environments identically.

5. Tags and cost control

Tag resources with owner and project, set budgets and alerts, and delete test resource groups when finished. Check current pricing and free offers on the Azure website, as they change.

Part C: Check yourself

- What is the difference between continuous delivery and continuous deployment?
- Name the three DevSecOps scans and one tool for each.
- Give one example each of IaaS, PaaS and SaaS on Azure.
- When would you choose rehost over rebuild?
- What happens when you delete a resource group?

Part D: Practice ideas for tonight

- Run bandit on your SmartDesk back end and fix one finding.
- Write an azure-pipelines.yml with Build and Security stages for your Day 15 tests.
- Classify ten tools you use at work as IaaS, PaaS or SaaS.

Coming up on Day 18: context engineering and RAG foundations.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today